



Abhijeet Kumar, Founder



Abhinav, Founder

ent categories has been designed in such a way as to prevent thermal runaway, hence preventing any chance of a fire. The company's custom-designed BMS on a PCB can be used for any electric two-wheeler, three-wheeler, or four-wheeler, or even a single solar cell or an entire solar panel.

The company has designed ten different BMS systems for different types of lithium-ion batteries, ranging from lithium iron phosphate to lithium cobalt oxide. The company takes pride in being able to customise its battery management system as per a customer's voltage and current requirements.

The company has secured a pat-



Body Disinfectant Machine

ent for their innovation. Commenting on the achievement, Founder Abhijeet Kumar says, "For every BMS to function properly, we need to control the temperature, voltage, and current. In addition to that we need to optimise the losses in the current and voltage rating. That is where we are different. Our sleeping current is very less. In ideal conditions, our sleeping current is less than $6\mu\text{A}$."

Incubated at IIT Patna, the founding team got together in August 2018 to build a lithium-ion battery and BMS specifically for defence applications. Supported by DRDO and IIT Kanpur, who are among its customers, the company is set to commercialise the product from March 2023.

DRY DISINFECTION

The Covid-19 pandemic exposed us to the lax healthcare system in our country. While our doctors and nurses broke their backs trying to save lives, many Covid warriors lost their lives after coming in contact

with the disease due to a lack of proper protection kits.

Amid these happenings, a young couple with no technical background, but a will to save lives, started Kingshahi Innovations Pvt Ltd. Startup entrepreneurs, Barun Shahi and his wife Meenakshi Kumari, have come up with a full-body disinfectant machine to completely disinfect any human in hospitals.

As the name suggests, the machine automatically deploys a disinfectant solution in the form of micro-droplets of the size of $50\mu\text{m}$ when its electronics sensors detect the presence of an adult human being, weighing not more than 125kg. An air pressure system on top of the machine prevents the micro-droplets from going beyond the shoulder level, avoiding any ingestion of disinfectant fluid by the subject.

The machine can currently work for two kinds of disinfectant fluids—ethanol based (70%) and sodium hypochlorite. It has successfully been tested to neutralise any